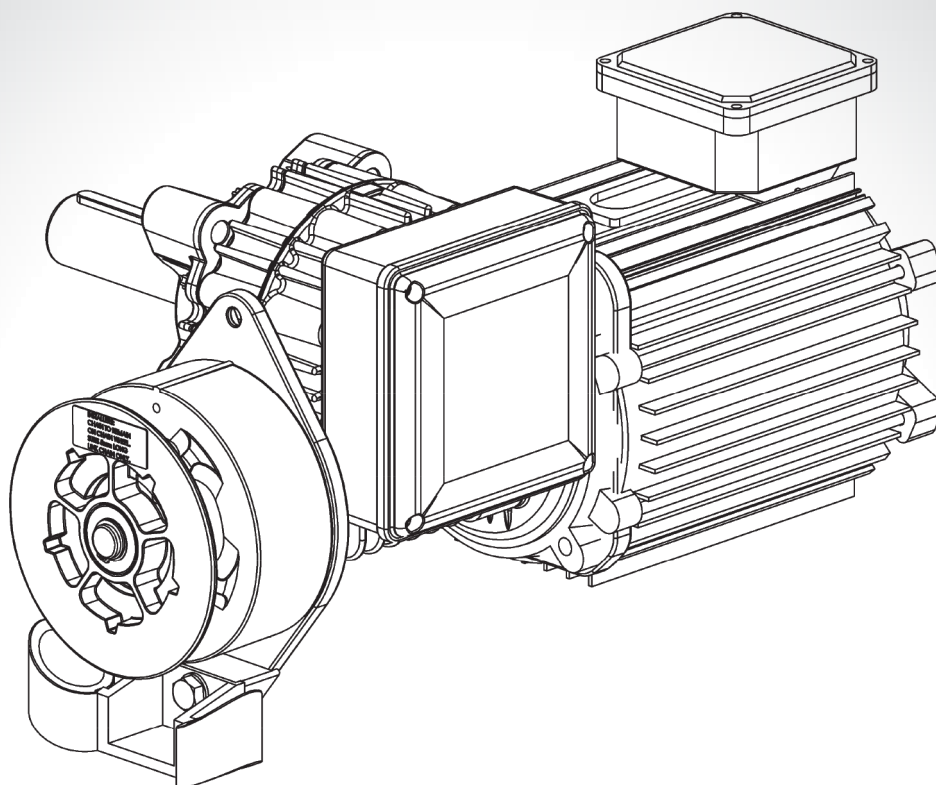


mDrive

INSTRUCTION MANUAL



mDrive Mechanical Door Operator

Installation and users manual for mDrive industrial door operator

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GRIFCO®

SAFETY INSTRUCTIONS



WARNING: THESE ARE IMPORTANT SAFETY INSTRUCTIONS.
FOLLOW ALL INSTRUCTIONS AS INCORRECT INSTALLATION CAN LEAD TO SEVERE INJURY OR DEATH
SAVE these instructions

This commercial roller door operator has been designed and tested to offer safe service provided it is installed, operated, maintained and tested in strict accordance with the instructions and warnings contained in this manual.



WARNING

Mechanical

When you see this Safety Symbol and Signal Words on the following pages, they will alert you to the possibility of serious injury or death if you do not comply with the warnings that accompany them. The hazard may come from something mechanical.

- Keep commercial door balanced. Sticking or binding doors must be repaired. Commercial doors, door springs, pulleys, brackets and their hardware are under extreme tension and can cause serious personal injury. Do not attempt to loosen, move or adjust them. Call for commercial door service.
- Do not wear rings, watches or loose clothing while installing or servicing a commercial door operator.
- To avoid serious personal injury from entanglement, remove all ropes connected to the commercial door before installing the door operator.
- After the installation a final test of the full function of the system and the full function of the safety devices must be done.
- When operating a biased-off switch, make sure that other persons are kept away.
- The operator cannot be used with a driven part incorporating a wicket door (unless the operator cannot be operated with the wicket door open).
- Operator may become hot during operation. Appropriate clearance and/or shielding should be supplied by the installer to ensure any cabling, wiring and/or other items cannot come in contact with the operator. If temperature rise exceeds 50°C all fixed wiring insulation must be protected, for example, by insulating sleeving having an appropriate temperature rating.
- Do not allow children to play with operator wall controls or remote controls. Keep remote controls away from children.
- Permanently fasten all supplied labels adjacent to the wall control as a convenient reference and reminder of safe operating procedures.
- Disengage all existing commercial door locks to avoid damage to commercial door. Install the wall control (or any additional push buttons) in a location where the commercial door is visible during operation. Do not allow children to operate push button(s) or remote transmitter(s). Serious personal injury from a closing commercial door may result from misuse of the operator.
- Activate operator only when the door is in full view, free of obstructions and operator is properly adjusted. No one should enter or leave the building while the door is in motion.
- The actuating member of a biased-off switch is to be located within direct sight of the door but away from moving parts. Unless it is key operated, it is to be installed at a minimum height of 1500mm and not accessible to the public.
- Make sure that people who install, maintain or operate the door follow these instructions. Keep these instructions in a safe place so that you can refer to them quickly when you need to.
- This appliance is not intended for use by persons (including children) with reduced physical, sensory or mental capabilities, or lack of experience and knowledge, unless they have been given supervision or instruction concerning use of the appliance by a person responsible for their safety. Children should be supervised to ensure that they do not play with the appliance.
- Use the operator for its intended purpose. The operator is designed to operate spring-balanced roller doors and roller shutters.
- Automatic Drive - Keep away from the area of the door as it may operate unexpectedly.
- Ensure that entrapment when operating the door in the open direction is avoided.



WARNING

Electrical

When you see this Safety Symbol and Signal Words on the following pages, they will alert you to the possibility of serious injury or death if you do not comply with the warnings that accompany them. The hazard may come from electric shock.

- If the supply cord is damaged, it must be replaced by the manufacturer, its service agent or similarly qualified persons in order to avoid a hazard.
- Installation and wiring must be in compliance with your local building and electrical codes. Connect the power supply cord only to properly earthed mains.
- Moisture and water can destroy the electronic components. Make sure under all circumstances that water moisture or storage moisture cannot penetrate the electronics. The same applies for openings and cable entries.
- An electrician must disconnect electric power to the commercial door operator before making repairs or removing covers.



CAUTION

When you see this Signal Word on the following pages, it will alert you to the possibility of damage to your commercial door and/or the commercial door operator if you do not comply with the cautionary statements that accompany it.

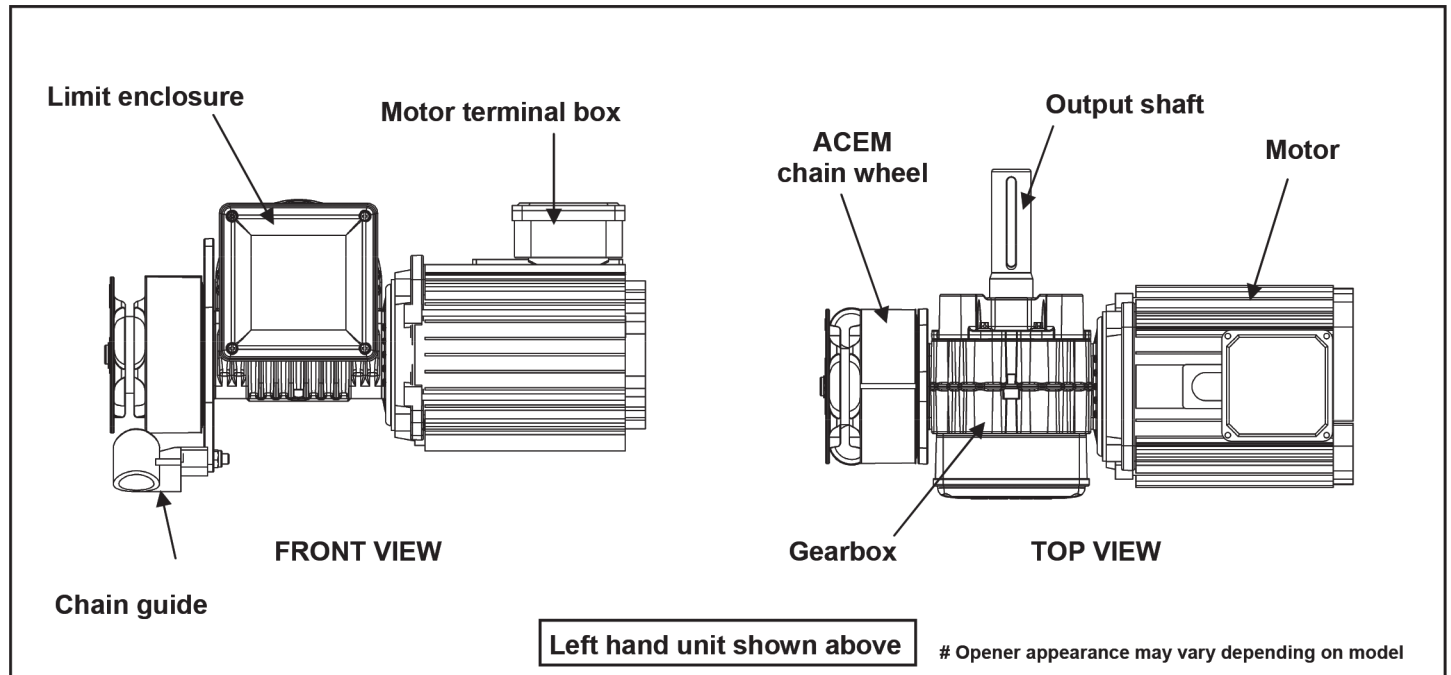
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INTRODUCTION

Congratulations on your purchase of the Grifco™ mDrive operator unit. For more than 50 years, we have been producing industrial mechanical operators that are well known for their proven reliability & robust design.

IDENTIFYING YOUR OPERATOR



WARNING MECHANICAL

Please remove any locks fitted to the door before operation in order to prevent damage to the door.

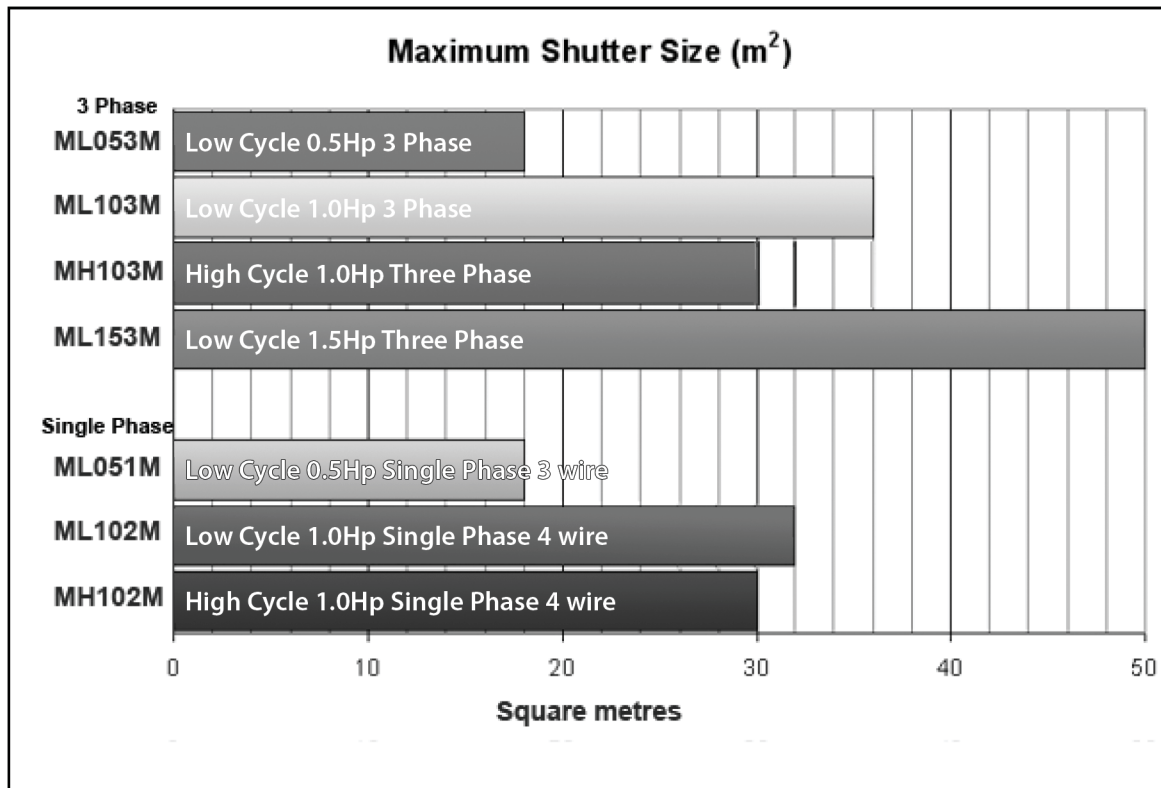


WARNING MECHANICAL

It is important to make sure the door always runs smoothly. Doors which stick or jam must be repaired immediately. Employ a qualified technician to repair the door, never attempt to repair it yourself.

PLANNING

Identify the type and dimensions of your commercial door. A survey of the application is recommended to see if any of the requirements below are not applicable with your installation.



Based on "standard shutter":

- The shutter is correctly spring balanced
- The shutter has $\leq 1.0\text{mm} \times \geq 100\text{mm}$ slats
- The shutter has no wind-locks
- A $\geq 4:1$ sprocket ratio is used

If any conditions above are not met, some consideration should be given to increasing the sprocket ratio, or operator size. The operator should be installed on the correct side of the commercial door. Consider an In-board mounting kit (P.No IBMK) if there is insufficient side room. Select the side that meets the requirements listed below.

- Must have minimum distance of 15mm between mounting plate and door drum sprocket
 - Must have minimum distance of 200mm between limit housing and imposing structure
 - Must have minimum clearance of 200mm from the terminal box
- Before installing the operator, check that the commercial door is in good mechanical condition, correctly balanced and open & closes properly.



CAUTION

The roller shutter guides must be fitted with mechanical stops that prevent the bottom rail from passing through in the opening direction. The operator should stall if driving into the mechanical stops.

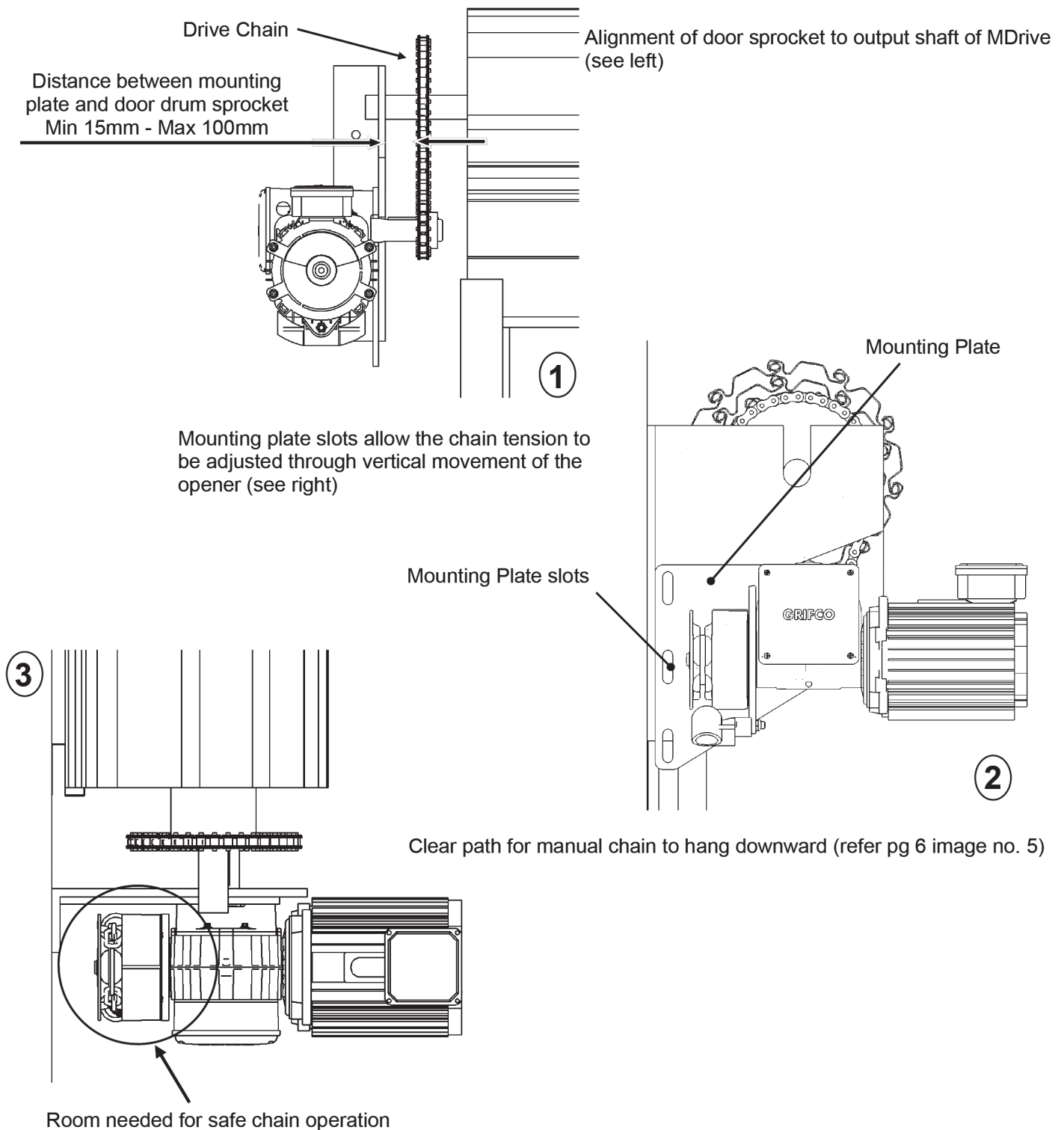
INSTALLATION

MOUNTING THE UNIT

The mDrive is typically flag mounted below the door drum so that the operator shaft points toward the door opening and lies beneath the sprocket of the door drum. For mounting you will need to either secure the operator to the roller shutter head plate with prepared holes or slots, or use a mounting plate that will need fixing via a wall angle or similar existing fixture.

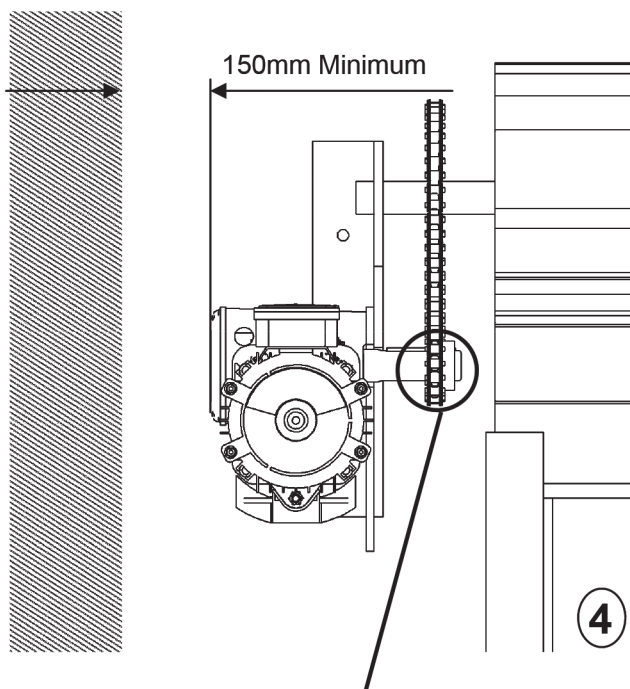
Note: THE mDrive IS NOT DESIGNED TO BE INSTALLED UPSIDE DOWN.
THE CHAIN GUIDE MUST NOT BE REPOSITIONED.

When assessing and selecting an appropriate mounting location, the following considerations should be made:



INSTALLATION

Mounting the Unit (continued)



Side room to imposing structures (see left)

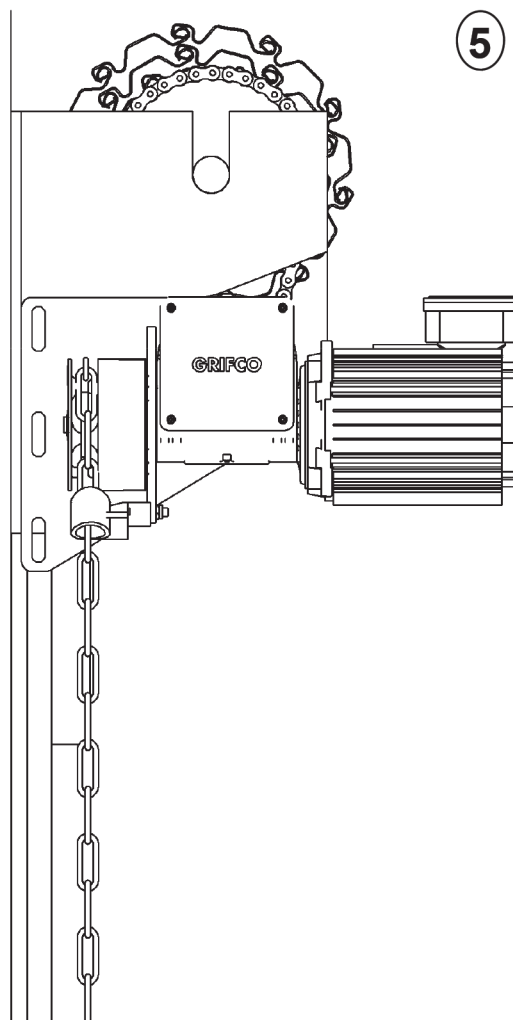
Note: The mDrive limit housing requires adequate clearance to allow setting of limit cams during installation. 150mm is the minimum recommended clearance between wall and lid. Where there is insufficient side room, consider using the opposite hand opener and mount in-board with a Grifco™ Inboard Mounting Kit, Part No. IBMK.

The drive sprocket must be aligned with the door sprocket, and locked using 2 x M8 hi-tensile grub screws tightened to min 20Nm. Grub screws must be set between 90° and 120° apart, with one grub screw lying over the drive key.

Chain tension should be set between 10-20kgs, ensuring both sides of the linkage are taught. Where practicable, use the weight of the opener as a preload, then tighten mounting bolts. M12 fasteners should be tightened between 80-90Nm.

Ensure the mounting position allows the hand chain to hang free of obstructions (see right)

Ensure hand chain is not twisted when making the join!



CAUTION

When Securing the mDrive Operator with the 4 x M12 fasteners (based on a 10mm mounting plate) and spring washers provided, it is critical to ensure that the applied torque is between 80-90Nm. When mounting through thicker sections, ensure a minimum of 30mm of screw thread is engaged with the female thread. Use of incorrect fasteners or torque may cause serious product damage and/or personal injury. When fixing through a slotted plate, ensure that the slots are no wider than 13mm as a spring washer may not be adequate in outside diameter to support the hexagon head.

INSTALLATION



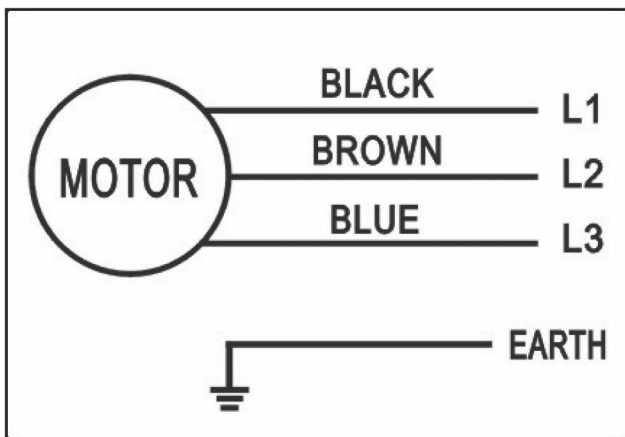
CAUTION

The mDrive Operator must:

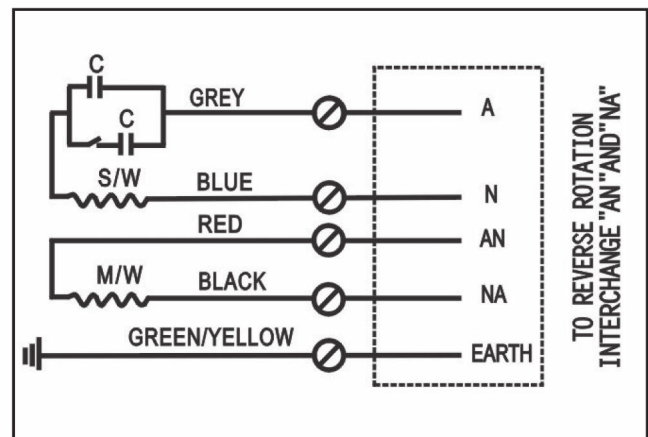
- be connected via a device that is capable of being locked for maintenance
- be connected via a suitable circuit breaker that disconnects all live conductors
- be connected in accordance with the wiring rules of the country in which it is installed
- not have control enclosures left open for extended periods (excess dust will void warranty)

Motor connections

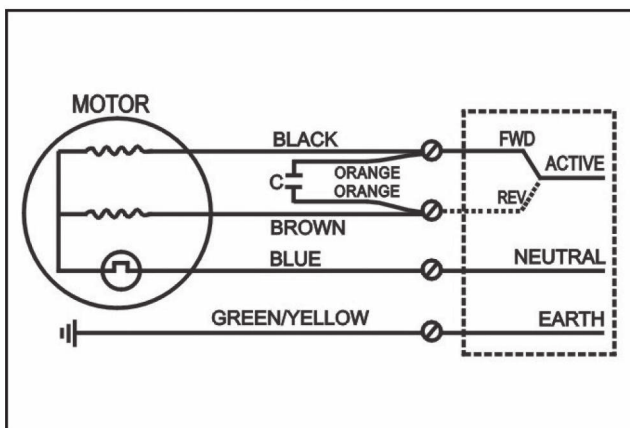
MODELS: Three phase 3 wire



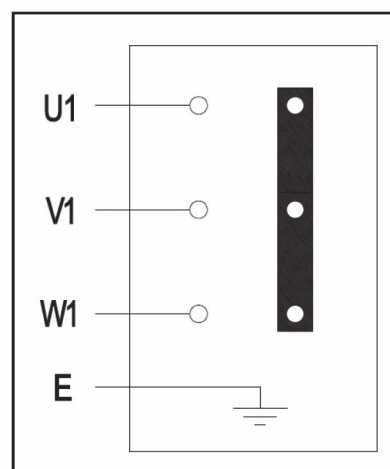
MODELS: Single phase 4 wire



MODELS: Single phase 3 wire



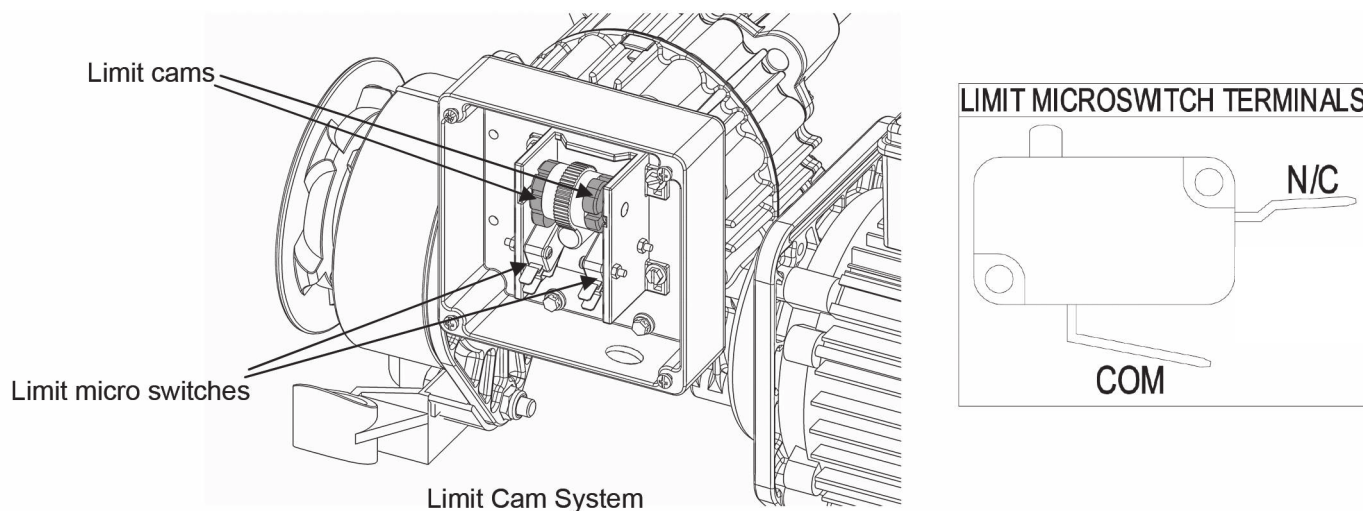
MODELS: To suit special 3ph openers



LIMIT SWITCH CONNECTION AND SETTINGS

CONNECTION

The limit switches must be connected to the control gear being used via the NORMALLY CLOSED (N/C) and COMMON (COM) of the microswitch:



The limit switch mechanism is a worm driven friction held adjustable cam system using two micro switches.



WARNING

Limits are extremely sensitive, a small cam movement may correspond to a large amount of shutter travel.



CAUTION

For operation with Reversing Starter controls refer to the RG wall control manual (D-M-ACG-033) for all set-up and safety instructions.

For operation with the GLM logic control refer to the GLM wall control manual (D-M-ACG-071) for all set-up and safety instructions.

LIMIT SETTING PROCEDURE

- ISOLATE MAINS BEFORE SETTING LIMITS
- AVOID CRAMMING WIRES INTO LIMIT ENCLOSURE AS IT MAY AFFECT CAM MOVEMENT

Follow the steps below:

1. Manually position the door in the centre of the opening.
2. Position one cam to depress a switch and the other so that it is well clear of the switch (note which switch it is that you have depressed).

For controls with up and down functions:

- Supply power and activate control in the upward direction.
- If the shutter travels up then the limit switch which is engaged by the cam is the bottom limit.
- If the shutter doesn't move it is the top limit.

For single button controls:

- Supply power and push the button.
- If the shutter moves up, the switch noted above is the bottom limit.
- If it moves down then it is the top limit.

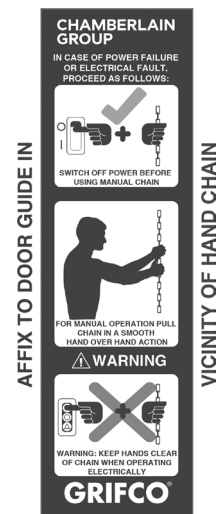
3. Isolate power and adjust the limit cams to depress the switches when the shutter reaches its open and closed Positions.
4. Test the cam positioning and adjust if necessary.

AUTOMATIC CHAIN MECHANISM

The patented ACEM features allows the use of the hand chain in the event of a power failure. Simply pull on the hand chain in either direction to operate the door manually.

Warning! Ensure power is isolated when using this feature.

The ACEM label (see right) must be fixed in close proximity to the controller e.g. on the door track.



GENERAL MAINTENANCE

MONTHLY

- Check chain alignment and tension. Adjust if required *
- Check PE / IR beam/s and bump edge functionality where applicable

QUARTERLY

- Check tightness of fixing bolts and (sprocket) grub screws. Adjust if required * refer page 6
- Check correct electrical operation
- Check manual operation via hand chain
- Lubricate the drive chain *
- Check door balance. Adjust if required *
- Conduct door maintenance in accordance with door manufactures guidelines. This will include door balance *

* SERVICE MUST BE CARRIED OUT BY A QUALIFIED TECHNICIAN

WARRANTY

Limited Warranty in Australia and New Zealand

1. Your consumer rights and guarantees

This Limited Warranty is provided by Chamberlain Australia Pty Ltd. Chamberlain New Zealand Limited (Chamberlain), contact details in Section 5 below. This Limited Warranty applies to a Grifco® Commercial unit (Unit) purchased in Australia and New Zealand, and gives you benefits which are in addition to your consumer rights and remedies under the Australian Consumer Law (or corresponding New Zealand consumer protection laws).

You can find out more information about your consumer rights and guarantees which the law provides in Australia at www.accc.gov.au, in New Zealand at www.consumerprotection.govt.nz. We also provide this statement as required under the Australian Consumer Law:

Our goods come with guarantees that cannot be excluded under the Australian Consumer Law. You are entitled to a replacement or refund for a major failure and for compensation for any other reasonably foreseeable loss or damage. You are also entitled to have the goods repaired or replaced if the goods fail to be of acceptable quality and the failure does not amount to a major failure.

2. What does our Limited Warranty cover?

Chamberlain warrants that, when purchased new in Australia or New Zealand, the Unit (all parts of the Unit other than globes and batteries) is free from defects in materials and workmanship (Limited Warranty) for the Limited Warranty period, subject to the terms and conditions of this Limited Warranty.

The Limited Warranty period (for Accessories see below) is 2 years (24 months) from the date of purchase or when a recommended cycle count for the purchased operator has been reached (whichever comes first).

The Limited Warranty period for remote controlled transmitters and accessories included with the Unit (Accessories) is 12 months from the date of purchase.

The Limited Warranty for genuine spare parts is free from defects in material and workmanship for a period of 6 months from date of purchase.

3. Limited Warranty Conditions

The following terms and conditions apply to your Limited Warranty:

our Limited Warranty is effective from date of purchase as indicated in Section 2 above;
proof of purchase of the Unit is required;

Please also see the User Manual for the Unit available on our website, or provided at the time of purchase.

4. What is not covered

Batteries and globes are not covered under the Chamberlain Limited Warranty

Travel costs incurred by Chamberlain or its authorised dealer in either travelling to and from areas outside a capital city area. These costs will be at the purchaser's expense.

Additional access costs incurred by Chamberlain or its authorised dealer in obtaining access to premises where the Unit is not readily accessible. These costs will be at the purchaser's expense.

Our Limited Warranty covers defects as explained, and does not cover all problems and mishaps that may occur in relation to the Unit including:

you got what you asked for but simply changed your mind, found it cheaper somewhere else, decided you did not like the purchase or had no use for it; you misused the Unit in any way that caused the problem; you knew of or were made aware of any deficiencies with the Unit before date of purchase;
use of the Unit with controls or third party devices or software which has not been supplied, or pre-approved, by Chamberlain;
problems relating to or residing in third party hardware, software or other items with which our product is used;
any loss of data related to you or provided by you, or loss related to downtime associated with use of the product whether through power outage, failure of internet or wireless connectivity, network disruptions, or otherwise;
non-compliance with the relevant instructions in the User Manual;
tampering, neglect abuse, wear and tear, accident, electrical storm, excessive use or conditions other than normal use;
problems with, or relating to, hardware, including but not limited to the

springs, hinges, guides:

problems caused by electrical faults or replacement of batteries or light bulbs, blown fuses, electrical surges, power surges or power strikes, fire, flood, rain, water, lightning or storms;
water or moisture ingress that causes corrosion or electrical malfunction; corrosion caused by sea air if located near a waterway, beach etc; lack of proper maintenance, service or care of the Unit; and/or damage caused by insects, pests or other after sale damage caused by events or accidents outside Chamberlain's reasonable control and not arising under normal and standard operating conditions.

5. Where you need help with our product

If you have a problem with the Unit or if you are concerned it may be defective, please contact our Customer Service team below:

Australia	Phone toll free 1800 474 326
Fax toll free	1800888121
Address	Unit 1, 75 Epping Road North Ryde, NSW 2113
Email	customerservice@chamberlainnz.com
Website	https://www.grifco.com.au/

New Zealand	Phone toll free 0800 653 667
Auckland	Phone 09 477 2823
Fax toll free	0800653 663
Email	customerservice@chamberlainnz.com
Website	https://www.grifco.co.nz/

As a first step, our Customer Service team will provide product support assistance to help you try to resolve the problem and in some cases, replacement parts for "do it yourself repairs". If our technical team need to see the product to determine the nature of the fault we will arrange an inspection by a trained technician, or ask that you send the product to us.

If your product was installed at your premises by an authorised dealer, please contact the installer indicated in your sales documents for prompt on-site service, or consult our Customer Service team if you need help in finding a local service technician. A service fee for on-site service may apply.

Authorisation from Chamberlain is required for warranty service under this Limited Warranty.

6. Delivery Charges

If an inspection is required in order to determine the nature of the fault please send the Unit freight paid and well packaged in accordance with our instructions. Any costs associated with transporting the Unit will be at Chamberlain's cost if the Unit has breached a statutory consumer guarantee. Repairs and replacement parts provided under this Limited Warranty are free of charge (and repaired items or replacement parts for Units found to be defective will be returned to you at our cost), and warranted by Chamberlain for the remaining portion of the original warranty period. However, if you make a claim under this Limited Warranty and this Limited Warranty does not apply, or if the law does not provide you with a statutory right or remedy as a consumer, you may be required to pay labour, assessment and/or freight costs and delivery charges to repair, replace and/or return the Unit to you.

7. How to register your product

Please register your Unit and contact details at this link <https://www.grifco.com.au/warranty-registration> so that we or your authorised installer can locate your details and provide you with product support assistance as quickly as possible, and keep you up to date with product updates. Registration can also be done by contacting Chamberlain Customer Service.

8. Important Reminder

All other guarantees (other than your statutory rights and guarantees as a consumer under applicable laws) are excluded to the fullest extent that we may lawfully do so. Unless the law requires us to do so, we do not accept any other liability, such as compensation for damages, injury or loss.

Last updated; 21st April 2022